Использование протокола STP. Агрегирование каналов

Лабораторная работа №9

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Цель работы

Цель лабораторной работы

Изучение протокола STP, механизмов отказоустойчивости сети, а также агрегирования каналов и перераспределения нагрузки.

Работа протокола STP

Резервное соединение

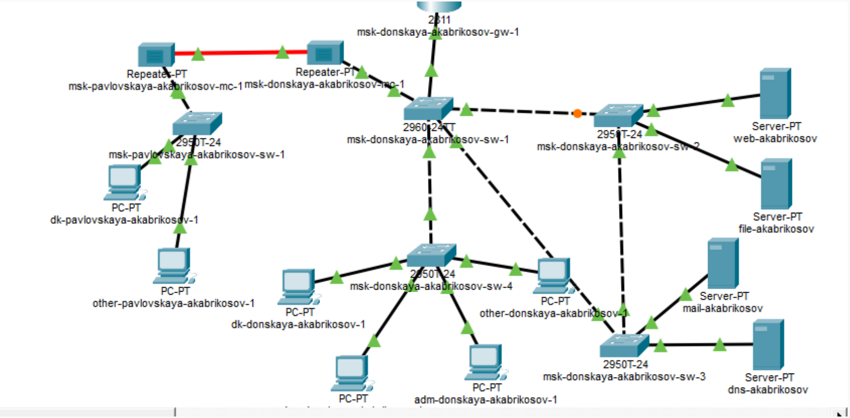

Рис. 1: Топология сети с резервным соединением

Проверка связности

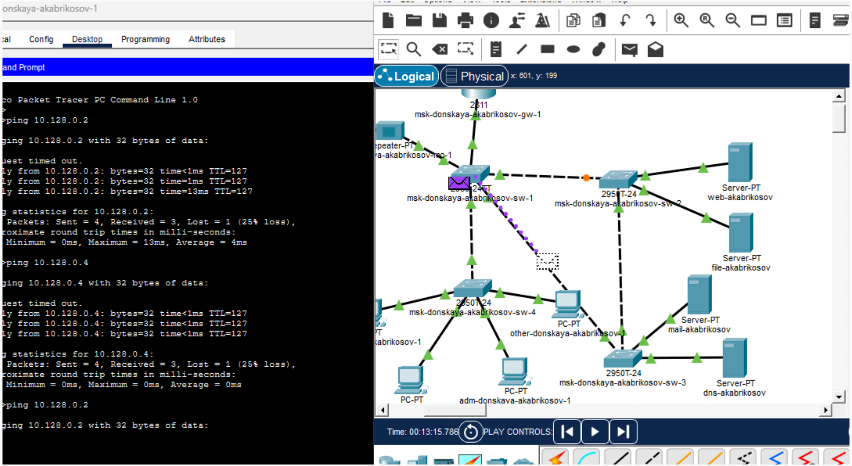

Рис. 2: Передача ICMP-пакетов

```
msk-donskaya-akabrikosov-sw-2#show spa
msk-donskaya-akabrikosov-sw-2#show spanning-tree vlan 3
VLAN0003
  Spanning tree enabled protocol ieee
  Root ID    Priority    32771
             Address     0001.97DE.99E8
             Cost        4
             Port        26(GigabitEthernet0/2)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32771  (priority 32768 sys-id-ext 3)
             Address     000D.BDE6.411E
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/1            Desg FWD 19        128.1    P2p
Fa0/2            Desg FWD 19        128.2    P2p
Gi0/1            Altn BLK 4         128.25   P2p
Gi0/2            Root FWD 4         128.26   P2p

msk-donskaya-akabrikosov-sw-2#
```

Рис. 3: Состояние STP VLAN 3

Настройка корневого коммутатора

Назначение root bridge

```
msk-donskaya-akabrikosov-sw-1(config)#spanning-tree vlan 3 root primary
msk-donskaya-akabrikosov-sw-1(config)#
msk-donskaya-akabrikosov-sw-1(config)#exit
msk-donskaya-akabrikosov-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-akabrikosov-sw-1#show spa
msk-donskaya-akabrikosov-sw-1#show spanning-tree vl
msk-donskaya-akabrikosov-sw-1#show spanning-tree vlan 3
VLAN0003
  Spanning tree enabled protocol ieee
  Root ID    Priority    24579
             Address     0003.E404.8D68
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    24579  (priority 24576 sys-id-ext 3)
             Address     0003.E404.8D68
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/23           Desg FWD 19        128.23   P2p
Fa0/1            Desg FWD 19        128.1    P2p
Gi0/2            Desg FWD 4         128.26   P2p
Fa0/24           Desg FWD 19        128.24   P2p
Gi0/1            Desg FWD 4         128.25   P2p

msk-donskaya-akabrikosov-sw-1#
```

Изменение маршрутов

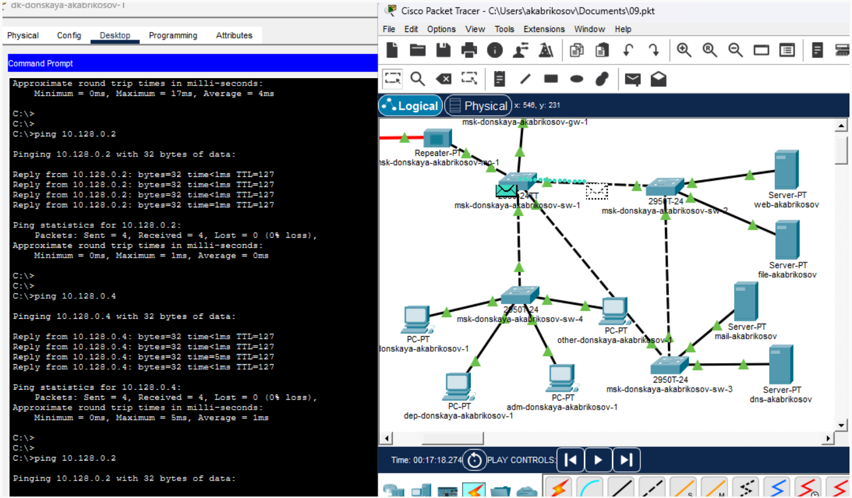

Рис. 5: Передача пакетов после настройки STP

Настройка PortFast

```
msk-donskaya-akabrikosov-sw-2(config)#int f0/1
msk-donskaya-akabrikosov-sw-2(config-if)#sp
msk-donskaya-akabrikosov-sw-2(config-if)#spa
msk-donskaya-akabrikosov-sw-2(config-if)#spanning-tree portf
msk-donskaya-akabrikosov-sw-2(config-if)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface  when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/1 but will only
have effect when the interface is in a non-trunking mode.
msk-donskaya-akabrikosov-sw-2(config-if)#int f0/2
msk-donskaya-akabrikosov-sw-2(config-if)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface  when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/2 but will only
have effect when the interface is in a non-trunking mode.
msk-donskaya-akabrikosov-sw-2(config-if)#exit
```

Рис. 6: Настройка PortFast

Отказоустойчивость STP

Потери пакетов при отказе

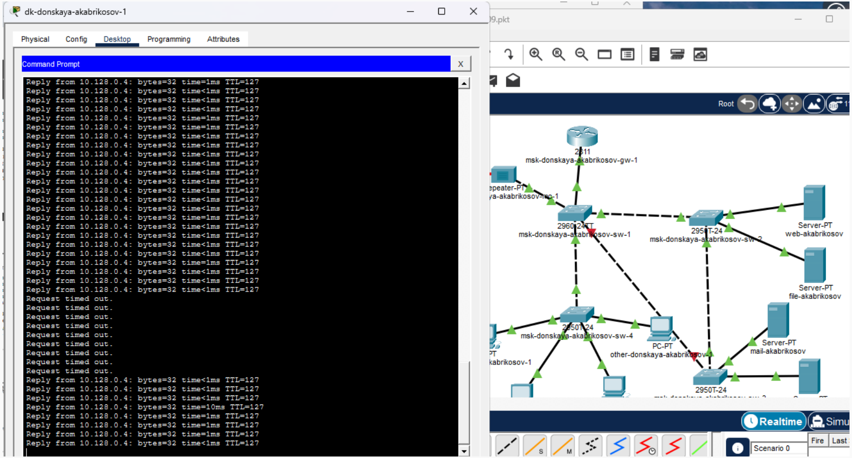

Рис. 7: Потери пакетов

```
msk-donskaya-akabrikosov-sw-1(config)#spanning-tree mode r
msk-donskaya-akabrikosov-sw-1(config)#spanning-tree mode rapid-pvst
msk-donskaya-akabrikosov-sw-1(config)#
msk-donskaya-akabrikosov-sw-1(config)#exit
msk-donskaya-akabrikosov-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-akabrikosov-sw-1#wr mem
Building configuration...
[OK]
msk-donskaya-akabrikosov-sw-1#
```

Рис. 8: Настройка Rapid PVST+

Работа сети

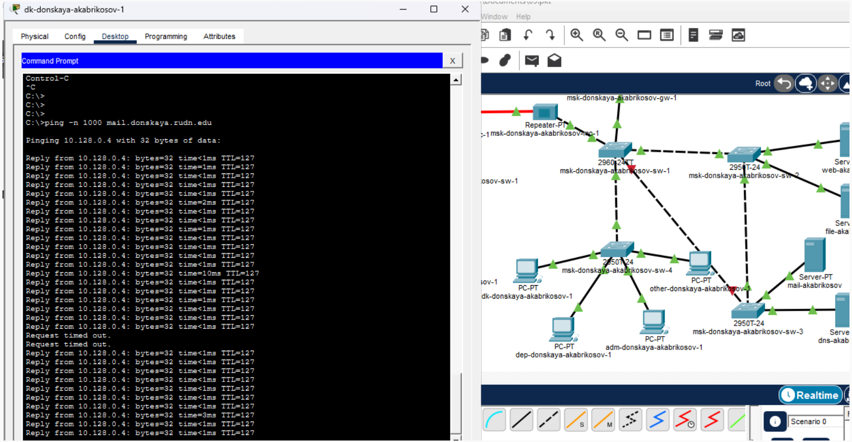

Рис. 9: Быстрое восстановление

Агрегирование каналов

Настройка Port-channel

```
msk-donskaya-akabrikosov-sw-1(config)#int range f0/20 - 23
msk-donskaya-akabrikosov-sw-1(config-if-range)#channel-group 1 mode on
msk-donskaya-akabrikosov-sw-1(config-if-range)#
Creating a port-channel interface Port-channel 1

%LINK-5-CHANGED: Interface Port-channel1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed state to up

%EC-5-CANNOT_BUNDLE2: Fa0/23 is not compatible with Fa0/20 and will be suspended (dtp
mode of Fa0/23 is on, Fa0/20 is off )

%EC-5-CANNOT_BUNDLE2: Fa0/23 is not compatible with Fa0/21 and will be suspended (dtp
mode of Fa0/23 is on, Fa0/21 is off )

%EC-5-CANNOT_BUNDLE2: Fa0/23 is not compatible with Fa0/22 and will be suspended (dtp
mode of Fa0/23 is on, Fa0/22 is off )

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23, changed state to down

msk-donskaya-akabrikosov-sw-1(config-if-range)#exit
msk-donskaya-akabrikosov-sw-1(config)#int port-channel 1
msk-donskaya-akabrikosov-sw-1(config-if)#sw mode trunk
msk-donskaya-akabrikosov-sw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23, changed state to up

msk-donskaya-akabrikosov-sw-1(config-if)#exit
msk-donskaya-akabrikosov-sw-1(config)#exit
msk-donskaya-akabrikosov-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-akabrikosov-sw-1#wr mem
Building configuration...
[OK]
```

Проверка ping

```
C:\>
C:\>ping 10.128.0.2

Pinging 10.128.0.2 with 32 bytes of data:

Reply from 10.128.0.2: bytes=32 time<1ms TTL=127
Reply from 10.128.0.2: bytes=32 time<1ms TTL=127
Reply from 10.128.0.2: bytes=32 time<1ms TTL=127
Reply from 10.128.0.2: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.128.0.4

Pinging 10.128.0.4 with 32 bytes of data:

Reply from 10.128.0.4: bytes=32 time=4ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.0.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 4ms, Average = 1ms

C:\>
```

Вывод

Итоги работы

- Изучен протокол STP и его поведение
- Исследована отказоустойчивость сети
- Настроен Rapid PVST+
- Реализован EtherChannel
- Повышена надежность и производительность сети